

Metabolomics Profiling after Allogeneic Hematopoietic Stem Cell Transplantation Unravels a Specific Signature in Human Acute GVHD

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Abstract

Introduction

Many efforts have been attempted to improve our understanding of GVHD pathophysiology but few focused on human GVHD. Most studies have focused on the central role played by immune cells. However, recent researches have highlighted the influence of tissue microenvironment or of host microbiota in regulating allo-immune response. In human, circulating metabolites are produced by both tissues and gut microbiota. Among them, many biochemical compounds regulate immune cells function and could be the bridge between microbiota, tissues and the immune system. The aim of this study was to characterize metabolomics alterations associated with allo-HSCT and to determine biochemical pathway specifically involved during acute GVHD.

Methods

We collected donors and recipients' frozen plasma from a monocentric (n=43) and a multicentric cohort (n=56) of patients who underwent transplantation from an HLA-identical sibling in France. Donors' samples were collected before stem cell collection. Recipient's samples were obtained at acute GVHD onset or at day + 90 in patients without GVHD. After protein removal, plasma extracts were analyzed by two separate reverse phase (RP)/UPLC-MS/MS methods with positive ion mode electrospray ionization (ESI), one for analysis by RP/UPLC-MS/MS with negative ion mode ESI and one for analysis by HILIC/UPLC-MS/MS with negative ion mode ESI. Compounds were identified by comparison to library entries of purified standards or recurrent unknown entities. Peaks were quantified using area-under-the-curve. All statistical analyses were performed with R 3.5.0 and the MetaboAnalystR package.

Results

More than 800 circulating metabolites were identified in the 2 cohorts, belonging to lipid (43.6%), amino acid (22.2%), xenobiotics (19.1%), nucleotide (4.2%), carbohydrate (3.6%), cofactors and vitamins (3.4%), peptide (2.8%) and energy (1.1%) pathways. After allo-HSCT, the comparison of metabolomics profiles of recipients without GVHD with those of healthy related donors revealed major changes in 222 metabolites (with a significance threshold at $p < 0.05$). SAM and Volcano Plot analysis (with a fold change (FC) > 2 and $p < 0.05$) highlighted significant modification in 43 metabolites. In the absence of acute GvHD, allo-HSCT was mainly characterized by a higher level of primary bile acids, mono and diacylglycerol, but a decrease of phospholipid, sphingolipid and secondary bile acid metabolites. Interestingly, polyamine, including N-acetyl putrescine (FC=5.5, $p < 0.001$) and N-acetyl spermidine (FC=3.7, $p < 0.0001$), were increased, suggesting that these microbiota-derived metabolites might play a protective role on gut integrity in patients without GVHD.

After comparison of recipients with or without GvHD, we were able to identify specific metabolomics changes associated with the onset of GvHD, irrespective of age, sex and feeding of patients. Acute GvHD was associated with a major decrease of plasmalogen and lysoplasmalogen (FC <0.5 , $p < 0.0001$), that may play a role in protection against oxidative stress induced by ROS. By contrast, medium and long chain fatty and polyunsaturated acid were strongly increased (FC >2 , $p < 0.01$). Most metabolites of amino acid pathways were decreased at onset of acute GvHD. Among them, Aryl Hydrocarbon receptor ligands were diminished in patients with GvHD, especially host- and microbiota-derived tryptophan metabolites such as 3-indoxyl sulfate ($p = 0.0001$), indole acetate ($p = 0.03$), indole propionate ($p = 0.03$) and N-acetyl kynurenine ($p = 0.04$). Many other gut microbiota-derived metabolites were significantly decreased in patients with GvHD, suggesting that major microbiota injury after allo-HSCT may regulate allo-immune responses through the production of metabolites with immunomodulatory properties.

Discussion

Our results demonstrate that allo-HSCT is associated with major metabolomics changes in recipients, which might be due to drug intakes, metabolic stress or microbiota alteration. GvHD was characterized by specific changes in complex lipid metabolism and amino acid metabolites that are involved in immune cells regulation and inflammation. This study highlights the potential role of circulating metabolites in GvHD pathophysiology that could be targeted for prophylaxis or treatment.

Figure

Disclosures

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Author notes

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